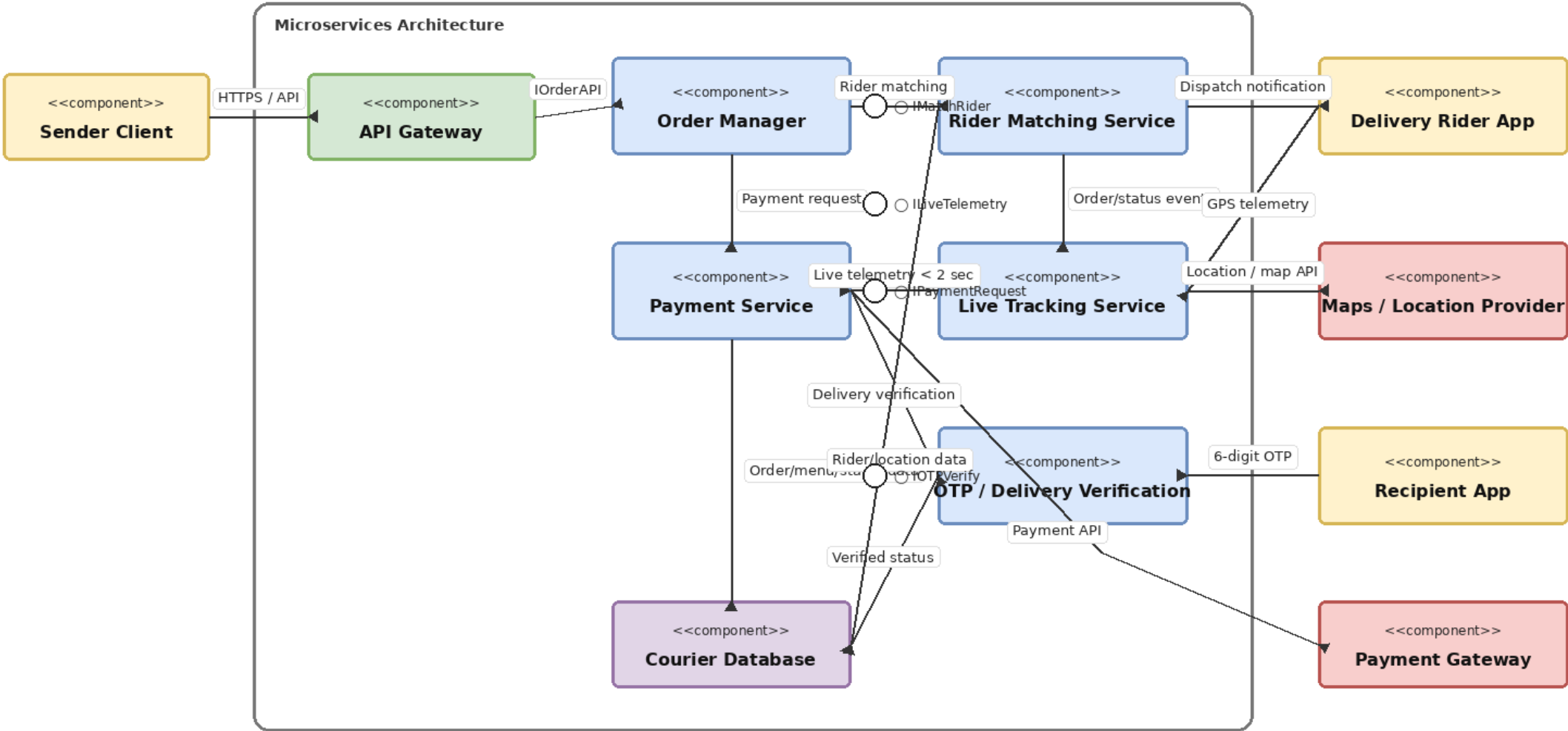


Hyperlocal Courier Dispatch & Tracking Engine

UML Component Diagram — Microservices Architecture



UML / Diagram Legend

○ Provided interface / service contract

→ Component interaction / data flow

Labels identify API, GPS, OTP, database and gateway flows

All labels are separated from connector lines for readability

Lab 1 Requirements Reflected

- Nearest active rider within 3 km
- Payment authorization before rider dispatch
- Live GPS telemetry under 2 seconds
- OTP required before Delivered
- 99.5% dispatch/matching availability
- Multi-stop routing supported by service layer